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Methodology of automated quality management

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Abstract: This research is devoted to the development of innovative technology for automation of the quality management system (QMS) of enterprises in Kazakhstan and its adaptation to the management system of enterprises. This article deals with quality, as an important strategic tool in business. System effectiveness evaluation of quality management enterprises is of great importance connected with the formation of rational decisions in the management of QMSs, including specificity of quality indicators, multi-level system, the necessity to choose the optimal number of performance indicators, and system status evaluation. The objective and relevance of this research are related to the need to (1) solve the problems of quality management in the digital economy, following from the relevant National programs of the Government of the Republic of Kazakhstan, which are important at this step of in-depth scientific research; (2) guarantee the competitiveness of domestic enterprises with high-quality requirements for products and services; (3) improve the efficiency of automated QMSs; (4) save resources (human and timing) in data processing. The method and model of automated enterprise quality management and intelligent automated system of quality management of enterprise integrated with ruling MICS (Management Information and Control System) subsystems are offered allowing to automate QMS implementation and support processes and increasing the validity, efficiency and effectiveness of management decisions by automating a number of functions of decision makers and personnel.

Keywords: quality management system, automated business processes, quality management, efficiency evaluation, fuzzy logic

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1 Introduction

Quality is an important strategic tool in business [1]. The goal of business process improvement is the transformation of the enterprise to meet the requirements of modern IT and management ideology in the aspect of the process approach. Problems of evaluating the effectiveness of the company's QMS are of great importance connected to the formation of rational decisions in the management of quality management systems (QMS), including specificity of quality indicators, multi-level systems, the necessity to choose the optimal number of performance indicators, and system status evaluation [2,3].

Information technology helps to change the relationship between consumption and production, and their interaction requires the exchange of information in order to organize and manage both manufacturers and consumers [4].

A monograph by Kubekov [5,6] has been studied as part of the ontological modeling of knowledge representation and management in the QMS, where the methodology of modeling knowledge components based on ontological engineering is presented, and new definitions are introduced. As to this project, we introduced a methodology for modeling business processes from the detailed engineering of architecture to the marketing of business logic [7,8].

Burkov et al. in Management Theory of Organizational Systems [9] discussed management models of organizational systems, as well as methods of solving management problems. The project [10] provides the basic mechanisms to manage organizational systems, and provides samples for the design of integrated management mechanisms, as well as mathematical models of the theory to manage organizational systems and their applications. We also devote attention to the complexity of computing the solution of problems of optimal management in the models of functioning of active systems, study effective methods of decision-making, and put management problems, using "parallelization" solution algorithms [11,12]. Conceptual and methodological research in management systems are presented in the literature [13].

The main objective for effective management is the stages of formation, implementation, and use of an automated QMS. The management system should provide

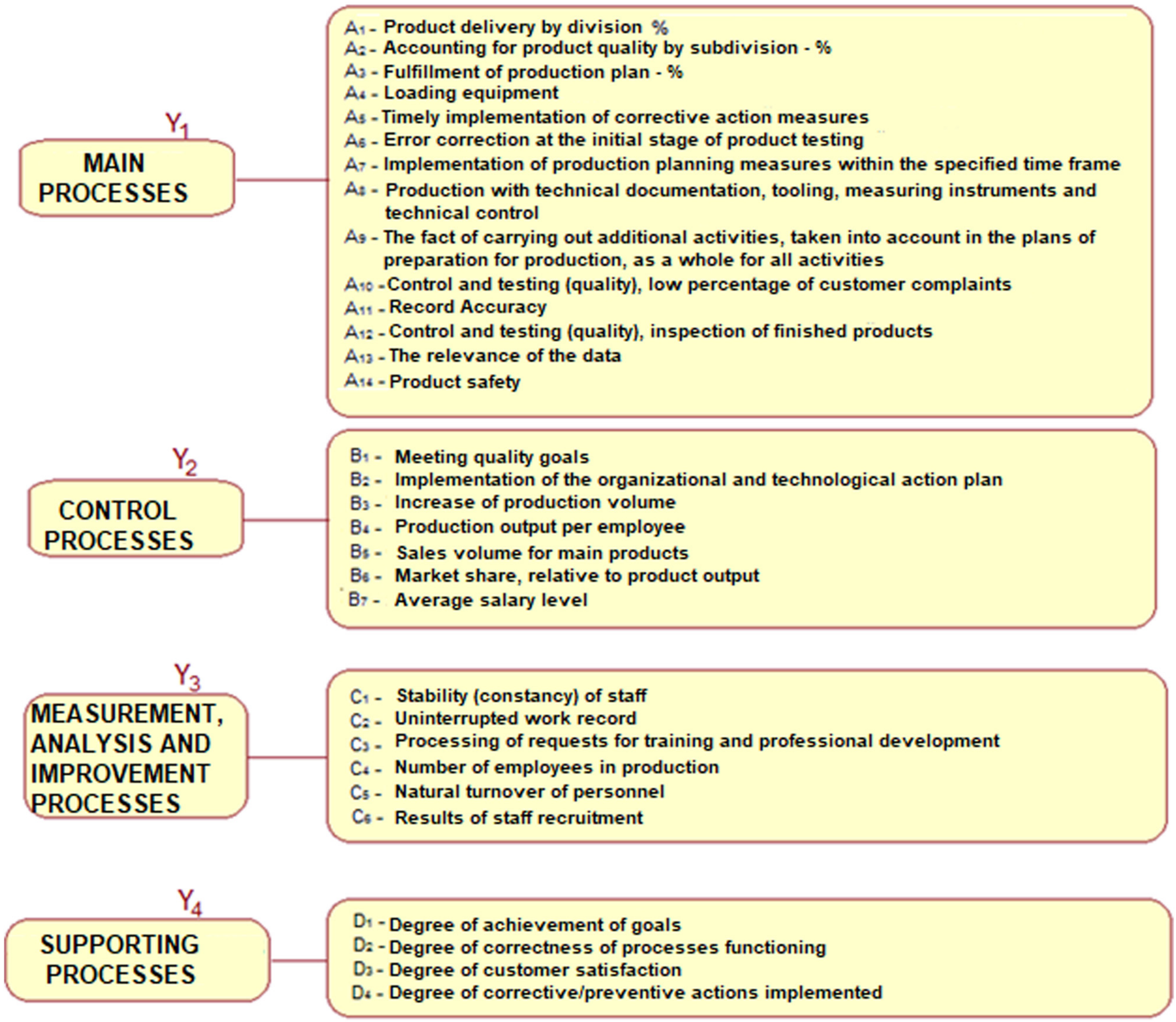

Figure 1: Business processes of the enterprise.

access to the documentation of the enterprise and qualitative conclusion of the requested information for receiving managerial operations to solve the problems of the enterprise as of a certain moment.

As to the analysis reports of the literature over the past 20 years, we can say that today in the world, especially in Kazakhstan, a scientific direction is developing related to the problems of implementation and automation of QMS, and technology analysis of business processes of various organizations.

2 The main part

It is necessary to solve the problems of introducing and maintaining up-to-date modern management systems in an industrially developed country, where competition, knowledge-intensive, innovative, and technologically sophisticated production are developed.

The Prime Minister of the Republic of Kazakhstan has adopted resolutions No. 28-r dated February 06, 2004, and No. 175-r dated April 27, 2006, in quality management due to the earliest rearrangement of enterprises according to ISO standards, and therefore, in order to achieve its objectives, the country is developing adequate logistics, regulatory and methodological system for the implementation of international standards [14,15].

The regulatory system of the Republic of Kazakhstan [14,16] includes 36 state standards, which are based on the international ISO standards and adopted as state standards of the Republic of Kazakhstan. Every year, the development and implementation of standards are included in

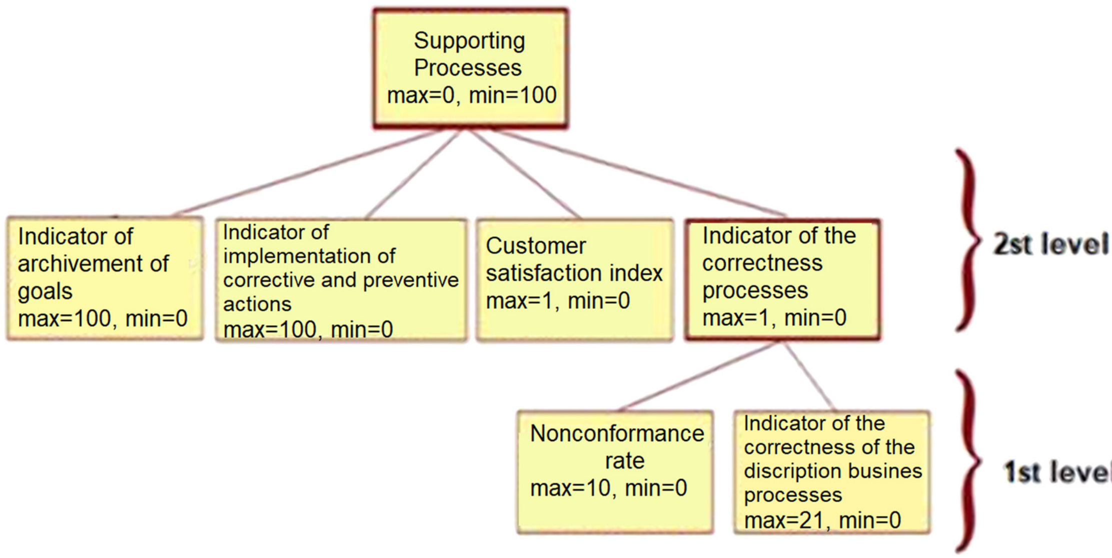

Figure 2: Two-level QMS assessment model.

the standardization plan of the Republic of Kazakhstan in management.

QMSs are key to maintaining the desired product quality and providing first-class services. QMS automate a wide range of business processes, including product design, standard operating procedures (SOP) development, management analysis, audits, training, claims management, corrective action preventive action (CAPA), etc. QMS users deal with huge amounts of data and various documents in their daily work. Handling such heterogeneous information manually can lead to human error and endanger products and consumers.

Table 1: I Rule database of the evaluation model QMS manual

No.	Inconsistencies	Correctness of description	Process
1	High	High	Low
2	Average	Average	Average
3	Low	Low	High
4	High	Low	Average
5	High	Average	Low
6	Low	Average	High
7	Average	Low	Average
8	Average	High	Low

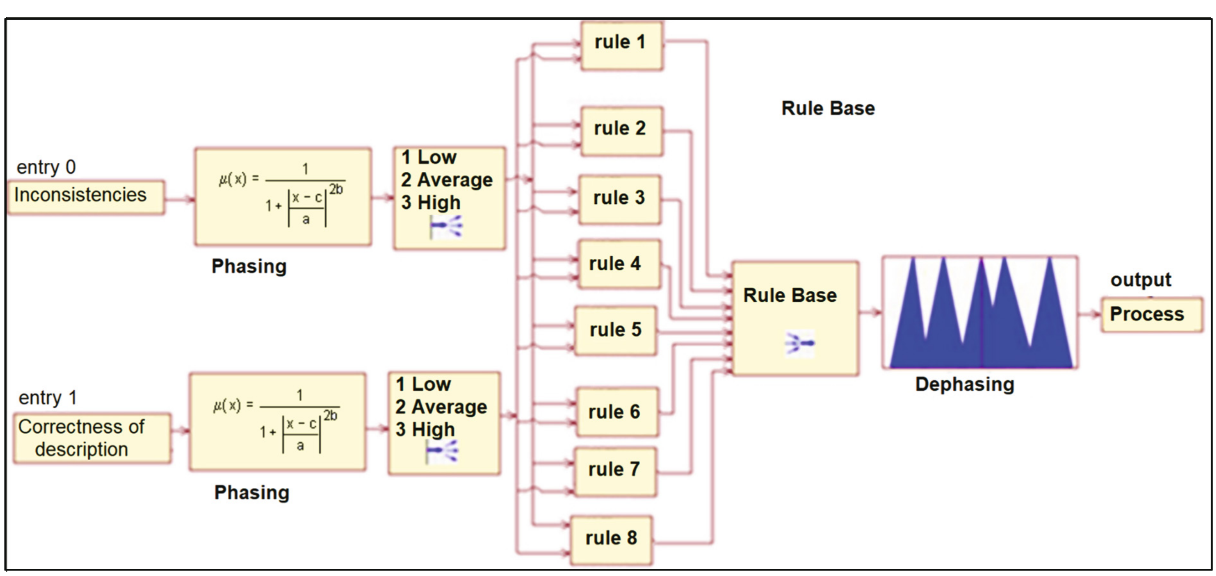

Figure 3: Fuzzy inference algorithm using the first level.

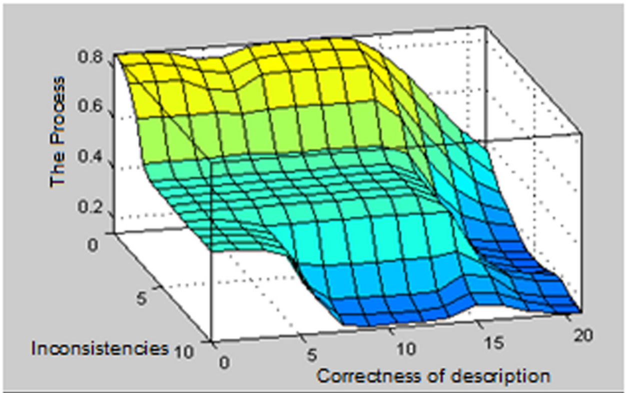

Figure 4: Matlab fuzzy model surface view window.

Table 2: The test result of the fuzzy output model level 1

No.	Inconsistencies	Correctness of description	Process	δ (%)
1	1	1	0.85	4.1
2	5	10.5	0.5	0.0
3	9	20	0.090	0.1
4	2	19	0.40	5.0
5	8	2	0.44	2.9
6	7	7	0.45	1.9
7	4	15	0.35	1.7
8	3	14	0.53	1.4

The automation of enterprises requires a lot of time and investment. Even if the system is designed and compliant with ISO standards, it will not provide assessment and prompt processing of a large amount of information related to the functioning of the organization. The information required is not communicated to a process on time for all

Table 3: Level-II rule database of the evaluation model QMS manual

No.	Achieving goals	Degree of proper functioning of processes	Customer satisfaction level	Degree of implementation of corrective/preventive actions	Performance assessment
1	2	3	4	5	6
1	Bad	Low	Not satisfactory	Low	1
2	Well (medium)	Low	Not satisfactory	Low	1
3	Perfect	Low	Not satisfactory	Low	2
4	Bad	Average	Average satisfactory	Average	2
5	Well (medium)	Average	Average satisfactory	Average	3
6	Perfect	Average	Average satisfactory	Average	4
7	Bad	High	Satisfactory	High	3
8	Well (medium)	High	Satisfactory	High	5
9	Perfect	High	Satisfactory	High	5
10	Bad	Average	Not satisfactory	Low	1
11	Bad	High	Not satisfactory	Low	2
12	Well (medium)	Low	Average satisfactory	Average	2
13	Well (medium)	High	Average satisfactory	Average	4
14	Perfect	Low	Satisfactory	High	3
15	Perfect	Average	Satisfactory	High	5
16	Bad	Low	Average satisfactory	Low	1
17	Bad	Low	Satisfactory	Low	1
18	Well (medium)	Average	Not satisfactory	Average	3
19	Well (medium)	Average	Satisfactory	Average	4
20	Perfect	High	Not satisfactory	High	4
21	Perfect	High	Average satisfactory	High	5
22	Bad	Low	Not satisfactory	Average	1
23	Bad	Low	Not satisfactory	High	1
24	Well (medium)	Average	Average satisfactory	Low	3
25	Well (medium)	Average	Average satisfactory	High	4
26	Perfect	High	Satisfactory	Low	4
27	Perfect	High	Satisfactory	Average	5
28	Bad	High	Not satisfactory	High	2
29	Bad	High	Average satisfactory	High	2
30	Bad	High	Satisfactory	Average	3

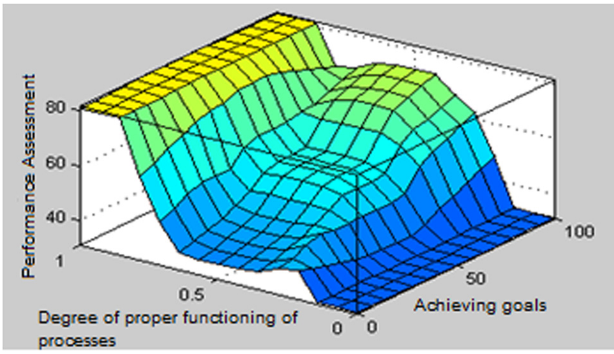

Figure 5: Level-II fuzzy model surface viewer window.

intents and purposes, and as a result, the decisions approved will largely not be fully adequate and are only addressed in an automated system [17].

The critically important fact is that quality management not only requires the use of automation tools but is also as well adapted as possible for their application. The provisions of the ISO 9000 series are based on the modification of the information flows of the enterprise [18], which makes possible the development and application of a running, in a way standard software.

Kazakhstan enterprises certify quality in their organizations as an important business strategy. Moreover, as information technology has developed, a problem arose in the obsolescence of traditional methods of data management on conformity of quality management [19,20].

Today, intelligent methods based on neural network technologies and fuzzy logic are used to solve management problems [21–24].

Artificial neural networks, based on learning and generalization algorithms, allow in some cases to successfully predict time series and reduce the requirements for mathematical training of subject experts, but neural network models cannot be formally imagined, and it is not possible to provide the results of time-series analysis.

The purpose of the article is the research and development of the methodology of automated management of the enterprise QMS, which allows us to automate the processes of implementation and maintenance of QMS at the enterprises of Kazakhstan and integrate with the existing subsystems of the automated enterprise management system for the implementation of the most important tasks arising from the State Program on digitalization of economic sectors.

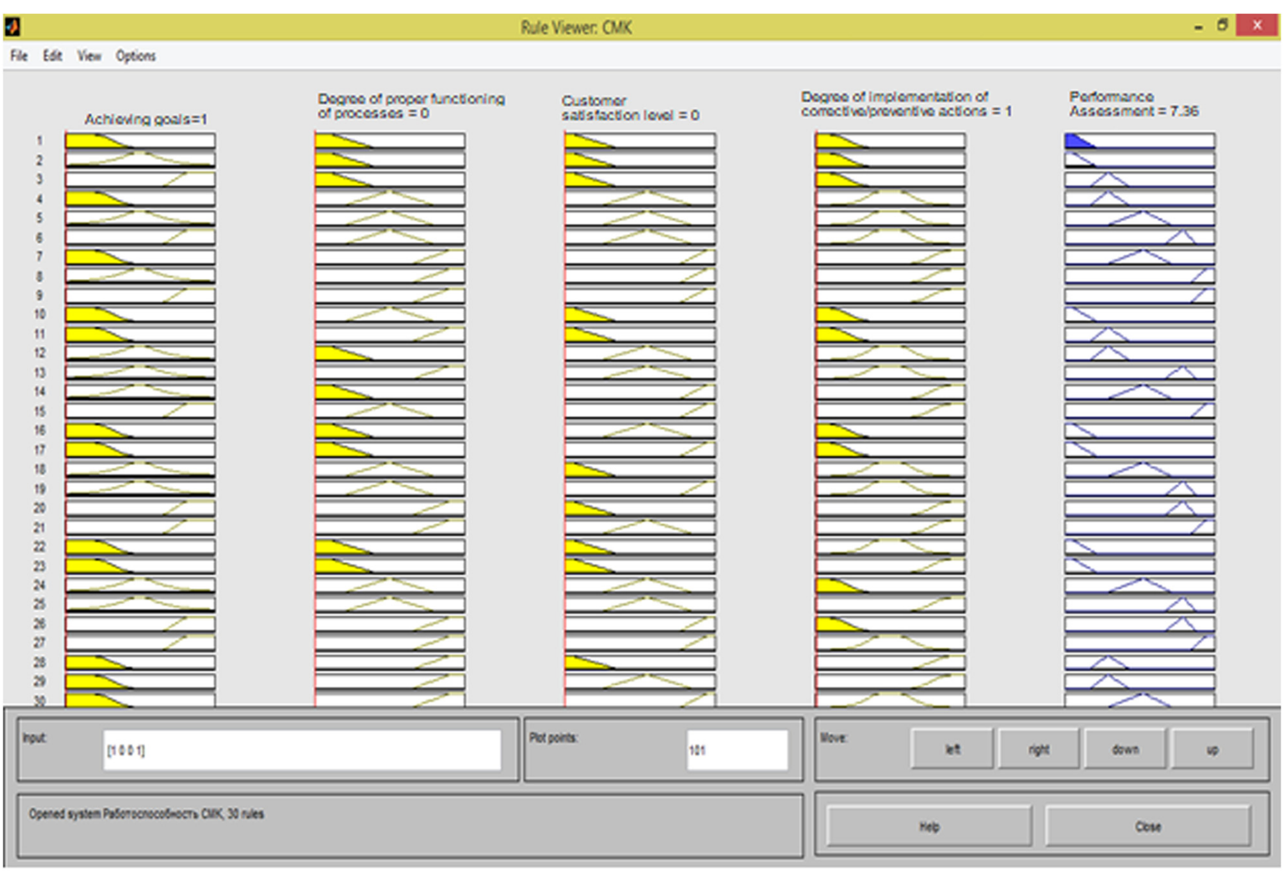

Figure 6: Level-II fuzzy inference rule viewer window.

Table 4: Model testing

No.	Achieving goals	Degree of proper functioning of processes	Customer satisfaction level	Degree of implementation of corrective/preventive actions	Performance assessment	δ, (%)
1	2	3	4	5	6	7
1	1	0	0	1	7.86	1.3
2	62	0	0	1	8.56	0.4
3	94	1	0	1	27.71	4.1
4	2	0.55	0.55	55	35.70	4.1
5	50	0.5	0.5	50	52.41	0.2
6	96	0.55	0.55	55	73.5	3.0
7	5	0.98	0.98	98	56.3	1.4
8	52	0.95	0.9	97	99.7	1.1
9	100	1	1	100	100	0.2
10	7	0.6	0.2	60	52.2	0.4
11	8	0.95	0.2	10	31	0.0
12	55	0.25	0.55	55	38.6	0.5
13	55	0.85	0.55	55	81.2	0.1
14	85	0.25	0.95	95	58.8	0.0
15	85	95	0.65	0.95	98	0.1
16	2	0.3	0.5	20	11.4	2.7
17	2	0.2	0.8	19	14.9	2.8
18	55	0.6	0.2	60	52.4	0.4
19	55	0.6	0.8	60	67.1	0.0
20	90	0.95	0.05	95	81.2	0.1
21	95	0.95	0.51	95	99.1	0.0
22	15	0.15	0.15	50	8.73	0.8
23	15	0.15	0.15	90	8.73	0.8
24	40	0.5	0.5	10	51.7	0.6
25	40	0.5	0.5	90	76.7	0.6
26	97	0.97	0.97	7	81.5	0.0
27	97	0.97	0.97	67	99.6	0.2
28	5	0.97	0.17	97	31.3	0.0
29	5	0.97	0.57	97	32.1	0.9
30	5	0.97	0.97	57	52.4	0.4

3 Discussion

In order to test the efficiency of the proposed methodology, models, and algorithms of automated quality management of the enterprise, a numerical study using the QMS data of the enterprise Innovation & Technologies LLP has been conducted. The data of the enterprise “Innovation & Technologies” LLP are considered received in the course of monitoring of processes by experts (heads of departments). Experts are specialists in the enterprise, and the number of experts is 6.

Intelligent management systems have developed over the past few years [25]. The main direction of development of these systems is the use of fuzzy logic apparatus: fuzzy set, fuzzy modeling, etc.

The fuzzy models of automated management systems are based on fuzzy logic controllers (FLCs) used to develop various automated process control systems (APCS), control systems for complex dynamic systems, etc. The FLC is based on fuzzy logic models: fuzzy link models and inference rules. The following system of linguistic description is popular for the FLC based on a fuzzy production processor: translation into fuzzy values (fuzzy value), fuzzy logical link, composite inference rules, and conversion operators into plain values (defuzzy value). The main step in designing an intelligent fuzzy controller is to create a “knowledge base” using representation methods and knowledge search.

Business processes of an enterprise are divided into several subgroups according to the ST RK ISO 9001 standard: Main processes, management (or managerial), and supporting (or otherwise auxiliary), and the number of processes depends on the specifics of the enterprise, as the standard does not define the exact number of processes but is only recommendatory (Figure 1).

The offered fuzzy model of production quality management gives an opportunity to predict indicators of

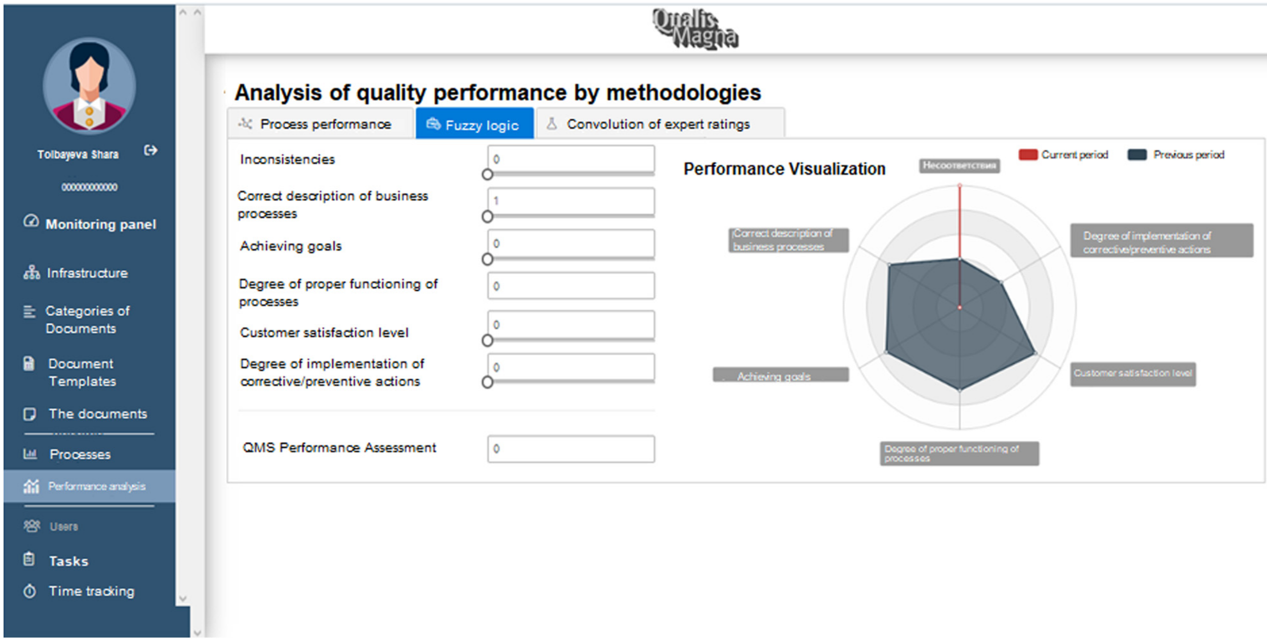

Figure 7: QMS progress analysis window.

the quality of services provided by the enterprise using "Supporting processes" for further use and introduction into the model.

Input and output variables of the two-level model are presented in Figure 2, where the output of the first level will be one of the inputs for the second level.

Figure 3 shows the fuzzy inference algorithm using the first level.

A Level-I output is an index of the correctness of business processes, the inputs are the variables, i.e. the index of inconsistencies (Inconsistencies) and the index of the correctness of the description of business processes (Figure 3). A sector is divided into three areas for variables, low, average, and high, defining their interval and membership functions.

Similarly, the rest of the enterprise QMS processes are calculated; thus, the possibility of obtaining quantitative estimates of processes in the developed model of intelligent quality management of production and business processes using the Mamdani fuzzy logic apparatus in the enterprise for QMS is shown. Using the program of viewing the surface of the fuzzy model, the adequacy of the constructed model and the influence of input variables on the output variable is proved.

Therefore, we will make a rule database, as shown in Table 1.

We can use the fuzzy model surface viewer shown in Figure 4 to find out the adequacy of the model, and as input variables affect the output one.

The test results of the created fuzzy inference model are presented in Table 2. As the systemic error δ (%) is not more than 5% to the original expert data, the developed model is considered adequate [p. 2.8, 118].

The rule database (a number of rules $T^{\text{input variable}} = 3^4 = 81$, after optimization = 30) is shown in Table 3.

We can use the Level-II fuzzy model surface viewer, shown in Figure 5, and determine the adequacy of the model, as input variables of Level II affect the output variable score (Figure 6).

Table 4 shows the model test results showing the positive effects of Level-II input variables on the output variable score.

Since the relative error does not exceed 5%, we can conclude that the developed rule bases are adequate.

Figure 7 shows the window of the automated quality management software developed by the authors of the enterprise QMS "Progress Analysis."

Figure 7 shows the window view to fill in the assessments of the company's QMS indices developed by the intelligent automated QMS of the enterprise.

4 Conclusions and follow-up

Effective data analysis is critical for upgrading and progressive improvement. Quality management technology provides data presentation in one place and facilitates data exchange and analysis. This can help to identify wasteful processes, quality defects, or inefficient equipment in less time.

Companies can improve quality processes in less time and take corrective actions with production management software. Overall results are better products and fewer customer complaints.

Using fuzzy logic theory for the analysis of QMS provides an opportunity to obtain fundamentally new models and methods for analyzing these systems [26].

It is reasonable to use the production form of knowledge stored in the QMS effectiveness assessment, which was confirmed in the development of a model of intelligent quality management of production processes using fuzzy logic apparatus [27].

The follow-up will focus on the development of the implementing models and algorithms for the digital transformation of documentation support for QMS to detect contradictions and inconsistencies in documentation support of QMS of Kazakhstan's Economy. This solution will eliminate the problems of processing high volumes of regulatory documents of the enterprise when accompanied by the automated QMS. The solution should be based on a reality model and be accompanied by the development of a formal language with approaches similar to the creation of well-known declarative programming languages.

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Conflict of interest: Authors state no conflict of interest.

Data availability statement: The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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